

SNIST 2027POs

Company Specific DSA PROBLEM STATEMENT SCHEDULE

Arrays, Strings, Linked List, Stacks, Queues, Trees, Graphs, Dynamic Programming, Backtracking and Greedy Algorithms

10-Day Schedule Overview

Day	Topic	Problem Count
Day 1 15-05-26	Arrays	7
Day 2 18-05-26	Strings	9
Day 3 20-05-26	Linked List	5
Day 4 22-05-26	Stacks and Queues	13
Day 5 25-05-26	Trees	6
Day 6 27-05-26	Graphs	6
Day 7 29-05-26	Dynamic Programming	6
Day 8 01-06-26	Backtracking	6
Day 9 03-06-26	Greedy Algorithms	6
Day 10 05-06-26	Mixed Practice / Assessment	12
Day 11 08-06-26	Grand Test	2Hrs

Detailed Topic-Wise Problem Statements

Day 1: Arrays - 3 hrs

Practice flow: Easy-level concept application followed by medium and hard-level placement-oriented problems.

S. No.	Problem Statement	Difficulty	Focus Area
1	MIN To MAX	Easy	Array basics and min/max logic
2	CodeChef Streak	Easy	Basic array traversal
3	Find the peak element in an array	Easy	Array traversal and comparison
4	Merge two sorted arrays	Medium	Two pointers and sorted arrays
5	Chef and the Magical Subarray	Medium	Subarray logic
6	Cops and the Thief Devu	Medium	Range marking and observation

GRADSTREET

1st floor, Phoenix towers, Genpact Rd, Uppal, Hyderabad, Telangana - 50007

7	Largest Common Element in Two Arrays	Hard	Searching, sorting or hashing
---	--------------------------------------	------	-------------------------------

Day 2: Strings - 3 hrs

Practice flow: Easy-level concept application followed by medium and hard-level placement-oriented problems.

S. No.	Problem Statement	Difficulty	Focus Area
1	Rotate String	Easy	String rotation
2	Longest Common Prefix	Easy	Prefix matching
3	DNA Storage	Easy	Encoding and string mapping
4	Largest Odd Substring	Medium	String scanning
5	Password	Medium	Character validation
6	Different Consecutive Characters	Medium	Adjacent character logic
7	Blobby Volley Scores	Medium	Simulation using strings
8	Lapindromes	Medium	Frequency counting
9	Minimum Insertions to Make Parentheses Valid	Hard	Parentheses balancing

Day 3: Linked List - 3 hrs

Practice flow: Easy-level concept application followed by medium and hard-level placement-oriented problems.

S. No.	Problem Statement	Difficulty	Focus Area
1	Remove Duplicates from Sorted List	Easy	Pointer traversal
2	Cycle in a Linked List	Medium	Fast and slow pointers
3	Delete the Middle Node of a Linked List	Medium	Two pointers
4	Find Intersection of Two Linked Lists	Hard	Pointer alignment
5	Linked List Increment	Hard	Carry handling and reversal/recursion

Day 4: Stacks and Queues - 3 hrs

Practice flow: Easy-level concept application followed by medium and hard-level placement-oriented problems.

S. No.	Problem Statement	Difficulty	Focus Area
1	Chef Builds Stack	Easy	Stack basics
2	Stack Sort	Medium	Stack-based ordering
3	Next Smaller Element	Hard	Monotonic stack
4	Convert Decimal to Binary	Easy	Stack or repeated division
5	Mountain Peak	Medium	Stack/array observation
6	Self-Destructing Strings	Medium	Stack string reduction
7	Chef Circular Search	Hard	Circular search logic
8	Stone Pile	Easy	Queue/simulation
9	Suspense String	Medium	Deque or string construction
10	Queue - To Do List	Medium	Queue operations

11	Sliding Window Maximum	Hard	Deque and sliding window
12	Necklace	Medium	Circular queue/string logic
13	Card Swipe	Medium	Queue/map simulation

Day 5: Trees - 3 hrs

Practice flow: Easy-level concept application followed by medium and hard-level placement-oriented problems.

S. No.	Problem Statement	Difficulty	Focus Area
1	Given a binary tree, find its height/depth and print the level-wise traversal.	Easy	Tree traversal and BFS
2	Check whether a given binary tree is a Balanced Binary Tree or not.	Medium	Height and recursion
3	Given a binary tree, print the left view and right view of the tree.	Medium	Level-order traversal
4	Find the Lowest Common Ancestor (LCA) of two given nodes in a Binary Tree.	Hard	Recursive tree logic
5	Given a Binary Search Tree, find the kth smallest element.	Medium	BST inorder traversal
6	Convert a sorted array into a height-balanced Binary Search Tree.	Hard	Divide and conquer

Day 6: Graphs - 3 hrs

Practice flow: Easy-level concept application followed by medium and hard-level placement-oriented problems.

S. No.	Problem Statement	Difficulty	Focus Area
1	Given an undirected graph, check whether it contains a cycle using DFS or BFS.	Medium	Cycle detection
2	Given a graph, perform BFS and DFS traversal from a given source node.	Easy	Graph traversal
3	Find the shortest path in an unweighted graph from a source node to all other nodes.	Medium	BFS shortest path
4	Given a directed graph, check whether it contains a cycle.	Hard	DFS recursion stack
5	Perform Topological Sorting of a Directed Acyclic Graph.	Hard	DAG and ordering
6	Count the number of connected components in an undirected graph.	Medium	DFS/BFS components

Day 7: Dynamic Programming - 3 hrs

Practice flow: Easy-level concept application followed by medium and hard-level placement-oriented problems.

S. No.	Problem Statement	Difficulty	Focus Area
1	Given an array, find the Longest Increasing Subsequence.	Medium	1D DP
2	Solve the 0/1 Knapsack Problem for given weights, values, and capacity.	Medium	Classic DP table
3	Given a set of coins, find the minimum number of coins required to make a given amount.	Medium	Unbounded DP
4	Find the Longest Common Subsequence between two strings.	Hard	2D DP on strings
5	Given a grid, find the minimum path sum from top-left to bottom-right.	Medium	Grid DP
6	Given a string, find the minimum insertions needed to make it a palindrome.	Hard	DP on strings

Day 8: Backtracking - 3 hrs

Practice flow: Easy-level concept application followed by medium and hard-level placement-oriented problems.

S. No.	Problem Statement	Difficulty	Focus Area
1	Solve the N-Queens Problem for a given value of N.	Hard	Backtracking with constraints
2	Given a maze, print all possible paths from source to destination using Rat in a Maze logic.	Medium	Grid backtracking
3	Generate all subsets of a given array.	Easy	Recursive choice model
4	Generate all permutations of a given string or array.	Medium	Swapping and recursion
5	Solve the Sudoku Solver problem using backtracking.	Hard	Constraint satisfaction
6	Given an array and a target, find all combinations whose sum is equal to the target.	Hard	Combination search

Day 9: Greedy Algorithms - 3 hrs

Practice flow: Easy-level concept application followed by medium and hard-level placement-oriented problems.

S. No.	Problem Statement	Difficulty	Focus Area
1	Given start and end times of activities, select the maximum number of non-overlapping activities.	Easy	Activity selection
2	Solve the Fractional Knapsack Problem using greedy strategy.	Medium	Sorting by value/weight ratio
3	Given train arrival and departure times, find the minimum number of platforms required.	Medium	Sorting and two pointers
4	Given jobs with deadlines and profits, schedule jobs to get the maximum profit.	Hard	Job sequencing
5	Given an array of intervals, merge all overlapping intervals.	Medium	Interval sorting
6	Given coin denominations, find the minimum number of coins needed to make a given amount using greedy approach.	Medium	Coin selection

Day 10: Mixed Practice / Assessment - 3 hrs

Practice flow: Easy-level concept application followed by medium and hard-level placement-oriented problems.

S. No.	Problem Statement	Difficulty	Focus Area
1	Find the peak element in an array.	Easy	Array traversal
2	Longest Common Prefix.	Easy	String prefix logic
3	Cycle in a Linked List.	Medium	Fast and slow pointers
4	Next Smaller Element.	Hard	Monotonic stack
5	Sliding Window Maximum.	Hard	Deque
6	Check whether a given binary tree is balanced.	Medium	Tree recursion
7	Detect cycle in a directed graph.	Hard	DFS recursion stack
8	Find the Longest Increasing Subsequence.	Medium	DP
9	Solve the 0/1 Knapsack Problem.	Medium	DP table
10	Solve the N-Queens Problem.	Hard	Backtracking
11	Find the minimum number of platforms required.	Medium	Greedy and sorting
12	Schedule jobs with deadlines and profits to maximize profit.	Hard	Greedy job sequencing

Topic-Wise Problem Count Summary

Topic	Easy	Medium	Hard	Total
Arrays	3	3	1	7
Strings	3	5	1	9
Linked List	1	2	2	5
Stacks and Queues	3	7	3	13
Trees	1	3	2	6
Graphs	1	3	2	6
Dynamic Programming	0	4	2	6
Backtracking	1	2	3	6
Greedy Algorithms	1	4	1	6
Mixed Practice / Assessment	2	5	5	12
Total	16	38	22	76

Final Outcomes

- Students will solve topic-wise problems from easy to hard levels with a clear progression model.
- Students will strengthen implementation skills in arrays, strings, linked lists, stacks, queues, trees, graphs, dynamic programming, backtracking and greedy algorithms.
- Students will improve problem identification skills by mapping each problem to its core technique such as traversal, two pointers, recursion, BFS/DFS, DP, greedy sorting, stacks or queues.
- Students will be ready for coding assessments, company-specific practice, and structured DSA revision.